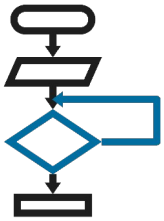
N5 Engineering Science
Programmable Control
Past paper questions and marking instructions



Contents

Question	What it covers	Marks	Answers
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2021 Question 13 (b)	Flowchart: rubbish compactor	11	Answers
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Sources and credits

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2021 Question 7

N5 Engineering Science 2021 paper, Section 1, Question 7. Total: 4 marks

MARKS | DO NOT WRITE IN THIS MARGIN

7. Microcontroller based systems are often used to create open loop and closed loop control.

(a) Describe the **difference** between open loop control and closed loop control. **2**

(b) Describe two advantages of a microcontroller based system compared to a hard wired electronic circuit. **2**

Advantage 1 _____

Advantage 2 _____

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2021 Question 7: marking instructions

Question			Expected response	Max mark	Additional guidance
7.	(a)		Open loop does not have feedback where closed loop has feedback.	**2**	Must be descriptive response. 1 mark for no feedback. 1 mark for has feedback. Do not accept has a sensor on its own.
	(b)		Fewer parts. Increased reliability. Simplified/faster assembly. Can be reprogrammed. Re-usable. Upgradable – system's features can be quickly/easily changed. Reduced stock inventory (one microcontroller circuit can be repurposed).	**2**	1 mark per each correct descriptive response. Accept – easier to fix mistakes. Not smaller or cheaper on its own. Not programmable.

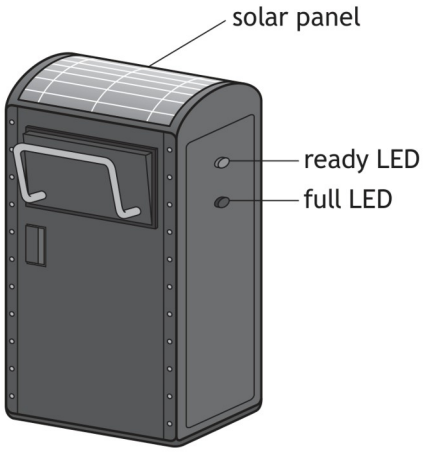
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2021 Question 13 (b)

N5 Engineering Science 2021 paper, Section 2, Question 13 (b). Total: 11 marks

MARKS | DO NOT WRITE IN THIS MARGIN

13. A solar powered rubbish compactor which crushes waste to increase its storage capacity is shown.



rubbish compactor

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MARKS | DO NOT WRITE IN THIS MARGIN

13. (continued)

A microcontroller is used to operate the rubbish compactor using the following sequence. The sequence runs after the rubbish compactor has been emptied and reset.

- A full LED will switch off and a ready LED will switch on
- When a rubbish level sensor is activated, a crusher will start to move down
- The crusher will stop when a force sensor is activated
- After a delay of 0·5 seconds the crusher will move up until a limit switch is activated
- If the rubbish level sensor is off, the sequence will return to the start
- If the rubbish level sensor is still activated the ready LED will switch off and the full LED will switch on
- The sequence will end

Input and output connections to the microcontroller are shown in the table below.

Input connection	Pin	Output connection
	7	crusher up
	6	crusher down
	5	full LED
	4	ready LED
force sensor	2	
limit switch	1	
rubbish level sensor	0	

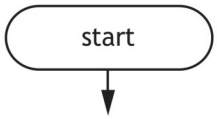
(b) Complete the flowchart opposite for this sequence, with reference to the data booklet and input/output connections.

Include **all** pin numbers and the delay unit in your flowchart. **11**

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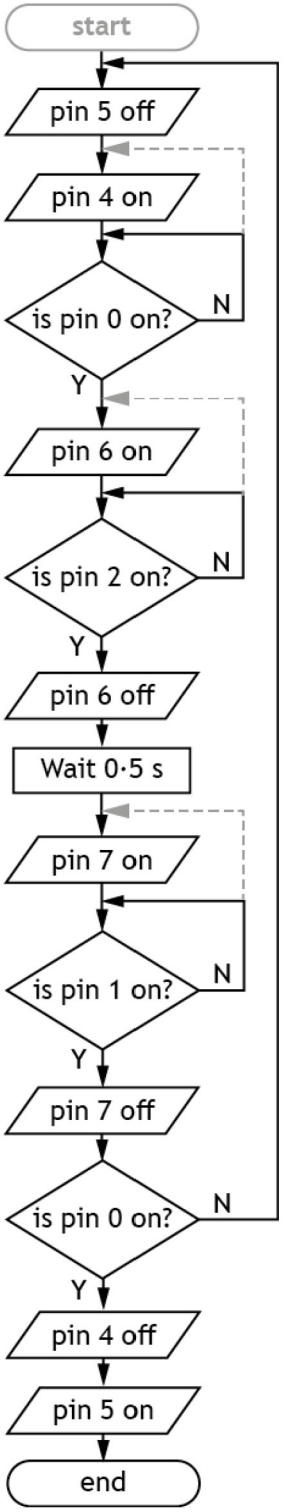
DO NOT WRITE IN THIS MARGIN

13. (b) (continued)



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2021 Question 13 (b): marking instructions

Question	Expected response	Max mark	Additional guidance
(b)		11	Pin numbers must be correct where applicable. Pin 5 off and on positions. **(1 mark)** Pin 4 on and off positions. **(1 mark)** Pin 6 on and off positions. **(1 mark)** Pin 7 on and off positions. **(1 mark)** 0·5 s delay. **(1 mark)** Pin 0 on ? with Y/N, loop and arrow. **(1 mark)** Pin 3 on ? with Y/N, loop and arrow. **(1 mark)** Pin 2 on ? with Y/N, loop and arrow. **(1 mark)** Pin 0 on ? (second) with Y/N, loop and arrow. **(1 mark)** End. **(1 mark)** All marked symbols correct. **(1 mark)** Ignore any additional steps.

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2022 Question 9

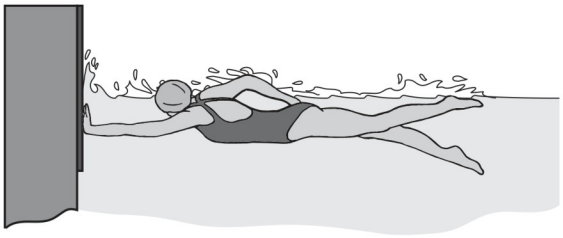
N5 Engineering Science 2022 paper, Section 2, Question 9. Total: 13 marks

DO NOT WRITE IN THIS MARGIN

SECTION 2 — 90 marks

Attempt ALL questions

9. In a swimming competition, a system is used to automatically measure a competitor's time.



The system is operated by a microcontroller.

The input and output connections to the microcontroller are shown in the table below.

Input connections	Pin	Output connections
	7	buzzer
	6	timer
lane switch	1	
master switch	0	

The system operates using the following sequence.

- A master switch is pressed
- A buzzer then sounds on and then off three times over 1.5 seconds
- The timer then starts
- When a lane switch is pressed the timer stops
- The system will then reset ready to be used again

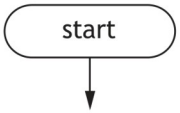
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MARKS | DO NOT WRITE IN THIS MARGIN

9. (continued)

(a) Complete the flowchart for the sequence, with reference to the data booklet and input/output connections.

Include **all** pin numbers and delay units in your flowchart. **10**



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MARKS | DO NOT WRITE IN THIS MARGIN

9. (continued)

A program used in a different control system is listed below.

```
line    program
1       main:     let count = 0
2       label_1:     switch on 4
3                    switch on 5
4                    pause 600
5                    switch off 4
6                    switch off 5
7                    pause 600
8                    let count = count + 1
9                 if count = 20 then label_2
10                goto main
11      label_2:  if Input0 is on then label_3
12                goto label_2
13      label_3:  switch on 7
14                pause 3000
15                switch off 7
16                goto main
```

(b) Describe the function of **line 16** in the program. **1**

_____

_____

Lines 2 to **9** should repeat twenty times before moving on to **line 11**.

During testing an electronic engineer found that this did not happen.

(c) Explain why **lines 2** to **9** did not repeat twenty times before moving on to **line 11**. **2**

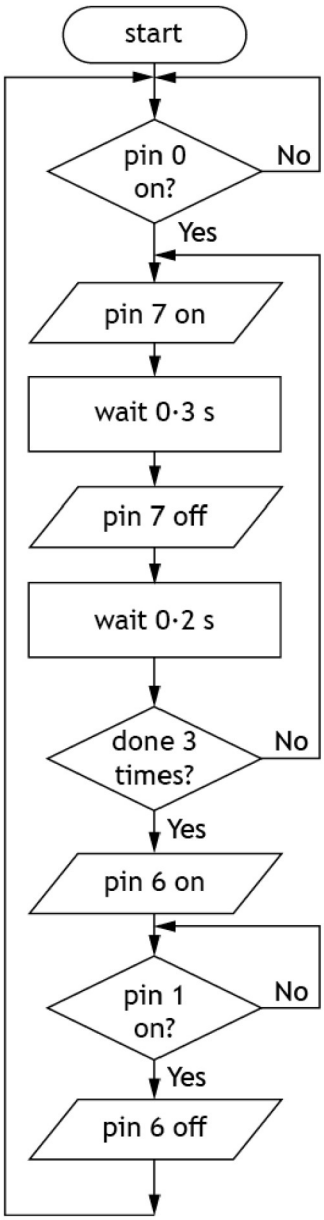
_____

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2022 Question 9: marking instructions

Question			Expected response	Max mark	Additional guidance
9.	(a)			10	Pin 0 on ? with Y/N, loop and arrow in correct position – 1 mark. Pin 7 on and off in correct position - 1 mark. Both delays in correct position (ignore incorrect values and unit) - 1 mark. Total delay time(s) with unit = 0.5 s per cycle - 1 mark. X3 decision with Y/N in correct position – 1 mark. Loop and arrow (x3 decision) back to before pin 7 on – 1 mark. Pin 6 on and off in correct position - 1 mark. Pin 1 on? with Y/N, loop and arrow in correct position – 1 mark. Accept decision feedback loop to before pin 6 on. Continuous loop to start with arrow - 1 mark. All marked symbols correct - 1 mark. Ignore any additional steps. accept on/off repeated 3 times max 2 marks.

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Question			Expected response	Max mark	Additional guidance
9.	(b)		To go back to line 1/main/restart the program To create a continuous loop.	**1**	Descriptive response. 1 mark for looping program back to start. Accept reset the program. Do not accept go to main on its own.
	(c)		The program loops back to line 1/main/"let count = 0"/wrong line ...therefore it will reset the count/ the count will not pass 1/count will not reach 20.	**2**	1 mark for program looping back to the line 1 (cause). 1 mark for resetting the count (effect).

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2022 Question 14 (c)

N5 Engineering Science 2022 paper, Section 2, Question 14 (c). Total: 1 marks

A microcontroller is used in the control of the hydropower station.

(c) Describe one **advantage** of using a microcontroller instead of a hard wired circuit. 1

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2022 Question 14 (c): marking instructions

Question			Expected response	Max mark	Additional guidance
	(c)		The program can be easily modified/ corrected It is reprogrammable Its more reliable because there are less components	1	Descriptive response relating to when in use. Do not accept cheaper/fewer components/reliable/quicker on its own. Do not accept easier/quicker to replace component.

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2023 Question 8 (a)

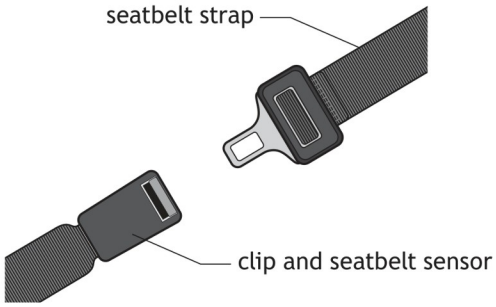
N5 Engineering Science 2023 paper, Section 2, Question 8 (a). Total: 10 marks

DO NOT WRITE IN THIS MARGIN

SECTION 2 — 90 marks

Attempt ALL questions

8. A car safety system detects when the driver's seatbelt is unfastened.



The safety system is operated by a microcontroller.

Input and output connections to the microcontroller are shown in the table below.

Input connections	Pin	Output connections
	7	warning lamp
	6	buzzer
seatbelt sensor	0	

The safety system operates using the following sequence:

- A warning lamp and a buzzer turn off.
- When the seatbelt sensor is on, the sequence will return to the start.
- If the seatbelt sensor is off, the warning lamp will turn on.
- The buzzer will then turn on and off three times over a total period of 1.8 seconds.
- The sequence will return to check the seatbelt sensor again.

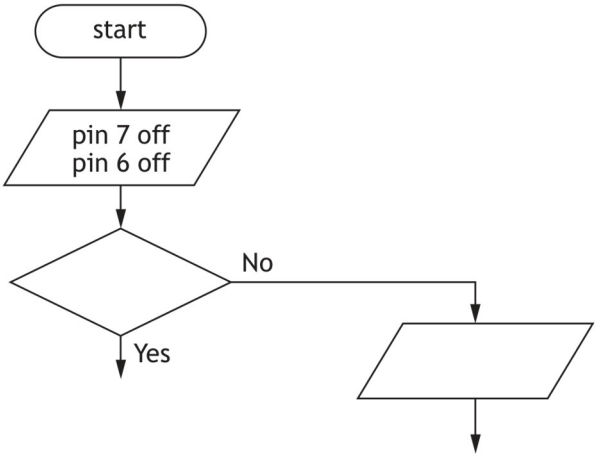
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MARKS | DO NOT WRITE IN THIS MARGIN

8. (continued)

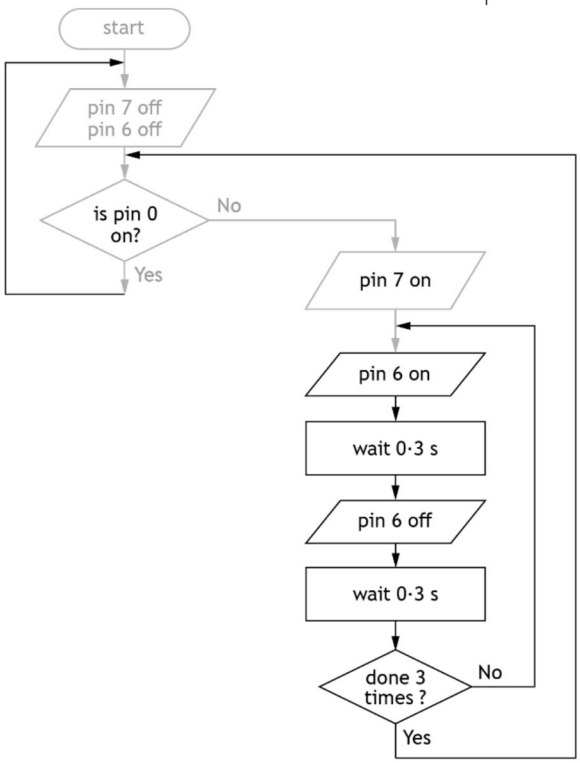
(a) Complete the flowchart for the sequence shown opposite, with reference to the Data Booklet and input/output connections.

Include **all** pin numbers and delay units in your flowchart. **10**



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2023 Question 8 (a): marking instructions

Question			Expected response	Max mark	Additional guidance
8.	(a)			10	Pin 0 on ? decision – 1 mark. Yes loop to start (arrow provided) – 1 mark. Pin 7 on in correct position – 1 mark. Pin 6 on **and** off in correct position – 1 mark. Both delays in correct position – 1 mark. Total delay time with unit = 0.6 s - 1 mark. x3 ? decision with Y/N – 1 mark. Correct repeat sequence loop with arrow back to before pin 6 on – 1 mark. Loop after sequence with arrow back to before pin 0 – 1 mark Accept (on, off, delay) sequence repeated three times – 2 marks. All symbols correct – 1 mark. Ignore any additional steps.

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2023 Question 9 (c) and (d)

N5 Engineering Science 2023 paper, Section 2, Question 9 (c) and (d). Total: 3 marks

MARKS | DO NOT WRITE IN THIS MARGIN

9. (continued)

A microcontroller based system is used to detect the bottles.

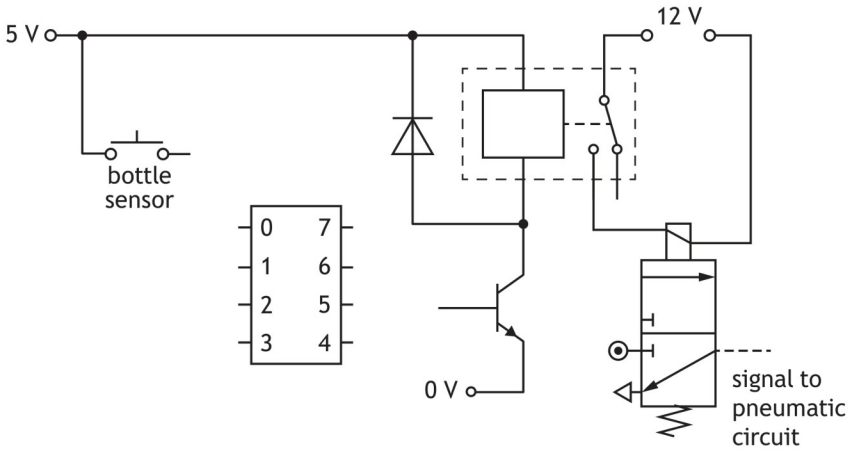
The program used to count six bottles is shown below.

line	program	
1	main:	let count = 0
2	check:	if input2 is off then check
3		let count = count + 1
4		if count = 6 then label_1
5		goto check
6	label_1:	switch on 7
7		pause 500
8		switch off 7
9		goto main

(c) State the **line number** that contains a time delay. **1**

An incomplete diagram for the microcontroller based system is shown below.

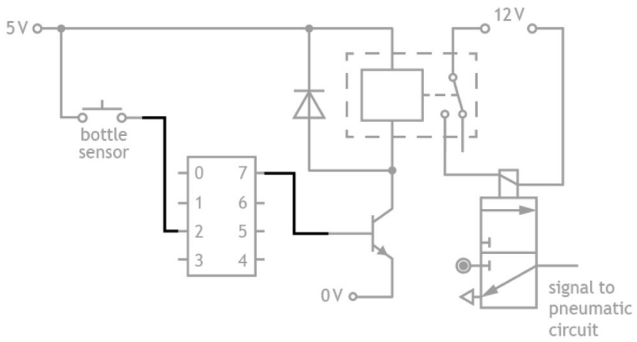
(d) Complete, with reference to the program above, the wiring of the bottle sensor and the transistor to the microcontroller. **2**



[Turn over

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2023 Question 9 (c) and (d): marking instructions

Question			Expected response	Max mark	Additional guidance
	(c)		(Line) 7	1	
	(d)			2	Bottle sensor correctly wired to pin 2 – 1 mark. Pin 7 correctly wired to base of transistor – 1 mark. Do not accept any additional pins wired to bottle sensor/transistor.

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2024 Question 11 (b)

N5 Engineering Science 2024 paper, Section 2, Question 11 (b). Total: 11 marks

MARKS | DO NOT WRITE IN THIS MARGIN

11. (continued)

The dishwasher is operated by a microcontroller.

Input and output connections to the microcontroller are shown in the table below.

Input connection	Pin	Output connection
	7	red LED
	6	water pump
	5	heating element
	4	buzzer
	3	
start switch	2	
door sensor	1	
	0	

A drying cycle used in the dishwasher operates using the following sequence:

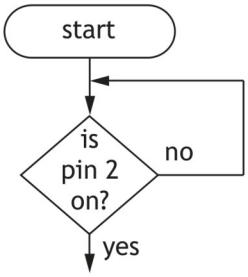
- When a start switch is pressed a red LED will turn on.
- After a door sensor is activated a water pump will turn on for 35 seconds and then turn off.
- A heating element will then turn on for 5 minutes and then turn off.
- A buzzer will then sound six times, turning on for 0.3 seconds and turning off for 0.3 seconds each time.
- The red LED will then turn off.
- The sequence will return to the start.

(b) Complete the flowchart shown opposite for the sequence with reference to the data booklet and input/output connections.

Include **all** pin numbers and delay units in your flowchart. **11**

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11. (b) (continued)

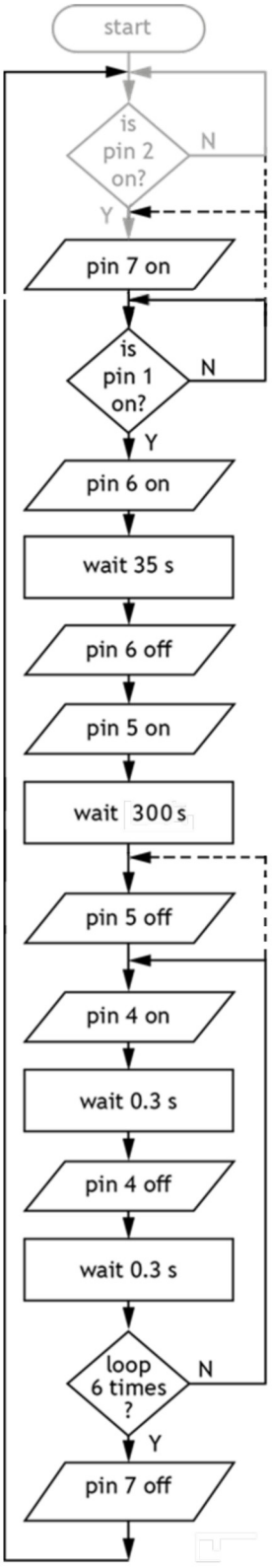


DO NOT WRITE IN THIS MARGIN

[Turn over

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2024 Question 11 (b): marking instructions

Question			Expected response	Max mark	Additional guidance
11.	(b)			11	Pin 7 on and off in correct position - 1 mark. Pin 1 on? with y/n - 1 mark. Pin 1 loop back with arrow before decision - 1 mark. Pin 6 on and off in correct position - 1 mark. Pin 5 on and off in correct position - 1 mark. Pin 4 on and off in correct position - 1 mark. All four delay times with correct units - 1 mark. Accept 5 minutes rather than 300 s. Loop x 6? with y/n - 1 mark. Feedback loop and arrow to before pin 4 on - 1 mark. Continuous loop and arrow to start of program - 1 mark. All symbols correct - 1 mark. Ignore any additional steps including their symbols.

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2025 Question 2

N5 Engineering Science 2025 paper, Section 1, Question 2. Total: 3 marks

MARKS | DO NOT WRITE IN THIS MARGIN

2. A microcontroller program is shown.

line	program	
1	`main:`	`if input1 is on then label_1`
2		`goto main`
3	`label_1:`	`let count = 0`
4	`label_2:`	`switch on pin 7`
5		`pause 500`
6		`switch off pin 7`
7		`pause 500`
8		`let count = count + 1`
9		`if count = 15 then label_3`
10		`goto label_2`
11	`label_3:`	`end`

(a) State a line number that contains a time delay. **1**

Line number _____

(b) State the number of times that pin 7 will turn on and then off. **1**

_____

(c) State the name of the type of loop used to control the number of times pin 7 turns on and off. **1**

_____

[Turn over

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2025 Question 2: marking instructions

Question			Expected response	Max mark	Additional guidance
2.	(a)		(Line) 5 or 7	**1**	Only a single line number required. Accept the line number indicated on the program.
	(b)		15 (times)	**1**	
	(c)		Fixed/ Finite (loop)	**1**	

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2025 Question 8 (c)

N5 Engineering Science 2025 paper, Section 2, Question 8 (c). Total: 2 marks

The pneumatic circuit is modified to be operated by a microcontroller.

(c) Explain why the use of a microcontroller is advantageous in this situation. **2**

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2025 Question 8 (c): marking instructions

Question			Expected response	Max mark	Additional guidance
	(c)		The microcontroller program can be easily updated/reprogrammed …without needing to change the pneumatic circuit. A microcontroller would need fewer (pneumatic) components ….reducing the cost of the circuit. The microcontroller program will allow a set number of piston movements …allowing accurate test results to predict performance.	**2**	Descriptive response relating to a positive reason for using a microcontroller in the context of a pneumatic circuit. 1 mark for cause. Do not accept easier to change on its own. 1 mark for effect.

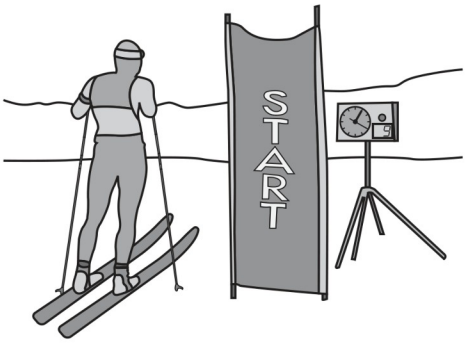
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2025 Question 9

N5 Engineering Science 2025 paper, Section 2, Question 9. Total: 13 marks

DO NOT WRITE IN THIS MARGIN

9. A skiing competition has each competitor starting the race with 1 minute intervals between them.



A microcontroller is used to control the starting signal for each competitor.

The sequence used is shown below.

1. A red lamp is turned on.
2. A start switch is activated.
3. The red lamp turns off and a buzzer turns on.
4. After a 0.5 second delay, the buzzer turns off and the red lamp turns on.
5. There is a 1 minute delay.
6. Steps 3 to 5 are repeated 50 times.
7. The red lamp turns off and the sequence ends.

Input and output connections to the microcontroller are shown in the table below.

Input connection	Pin	Output connection
	7	red lamp
	6	buzzer
start switch	1	

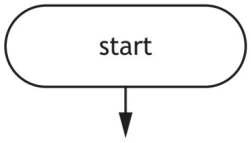
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MARKS | DO NOT WRITE IN THIS MARGIN

9. (continued)

(a) Complete the flowchart for this sequence, with reference to the data booklet and input/output connections shown opposite.

Include **all** pin numbers and the delay units in your flowchart. **10**

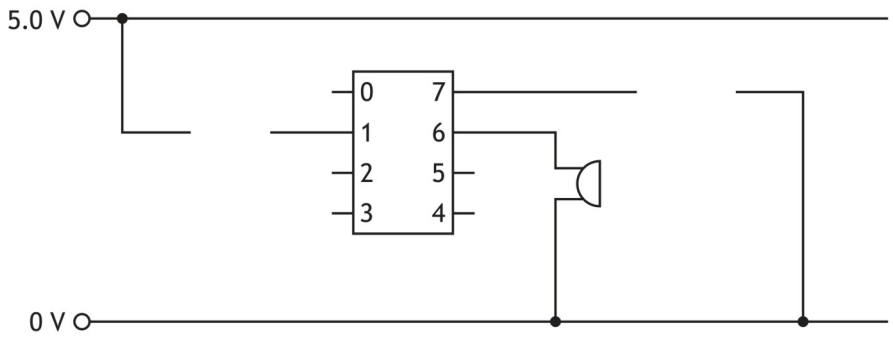


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MARKS | DO NOT WRITE IN THIS MARGIN

9. (continued)

An incomplete microcontroller wiring diagram used to test the sequence is shown below.



(b) Complete the wiring diagram with reference to the input/output table shown below. **2**

Input connection	Pin	Output connection
	7	red lamp
	6	buzzer
start switch	1	

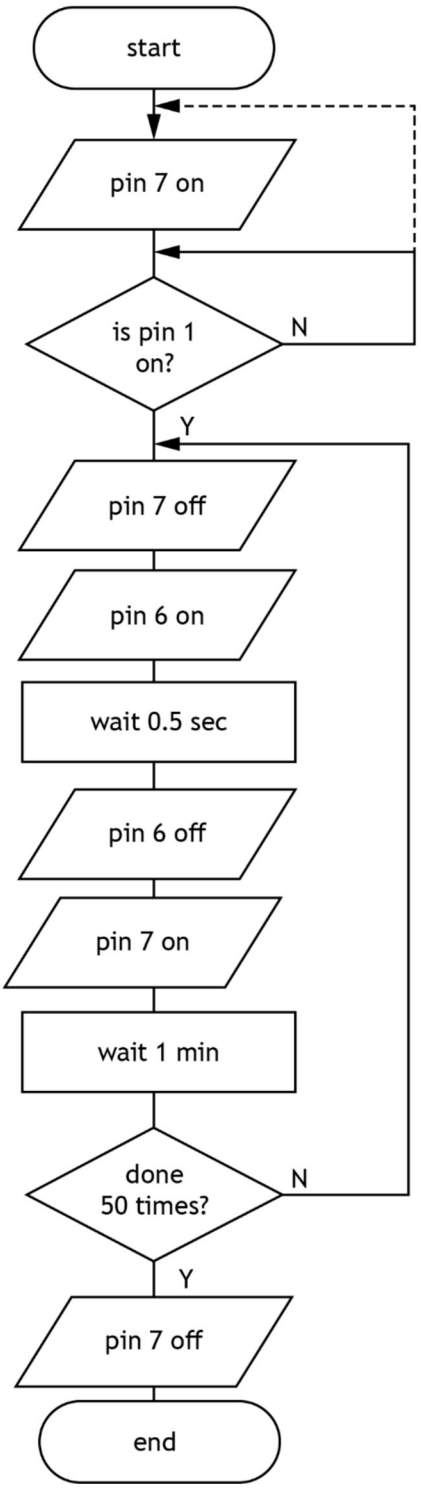
During testing it was found that the lamp was not bright enough.

(c) State the name of an electronic component that could be added to cause the lamp to turn on brightly when the pin is switched on. **1**

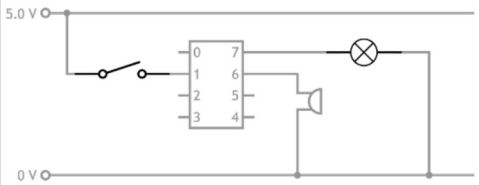
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2025 Question 9: marking instructions

Question			Expected response	Max mark	Additional guidance
9.	(a)			10	1 mark - (FIRST) Pin 7 on and off in correct position. 1 mark - Pin 1 on? with Y/N, loop and arrow in correct position. 1 mark - Pin 6 on and off in correct position. 1 mark - 0.5s delay in correct position with unit. 1 mark - (SECOND) Pin 7 on and off in correct position. 1 mark – 1 min (60s) delay in correct position with unit. 1 mark - x50 decision with Y/N in correct position. 1 mark - Loop with arrow (x50 decision) back to between pin 1 on and pin 7 off. 1 mark – end (stop). 1 mark - all correct symbols. Ignore any additional steps including their symbols.

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Question			Expected response	Max mark	Additional guidance
9.	(b)			**2**	1 mark for switch symbol wired between 5 V and pin 1. Accept push to make switch. 1 mark for lamp symbol wired between pin 7 and 0 V.
	(c)		Transistor	**1**	Accept relay (switch)/MOSFET. Do not accept driver.

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2026 Question 12 (a)

N5 Engineering Science 2026 paper, Section 2, Question 12 (a). Total: 10 marks

DO NOT WRITE IN THIS MARGIN

12. An electric toothbrush is shown.



The electric toothbrush is operated by a microcontroller.

Input and output connections to the microcontroller are shown in the table below.

Input connection	Pin	Output connection
	5	LED
	4	motor
battery level sensor (charged = 1)	2	
start switch	1	

The electric toothbrush operates using the following sequence:

- When the start switch is pressed, the battery level sensor is checked. If it is not pressed, the motor will be off.
- If the battery level is charged, the motor will turn on and the sequence will return to the start.
- If the battery level is not charged, the LED will turn on for 0.5 seconds and off for 0.4 seconds. This will repeat twenty times.
- The sequence will then return to the start.

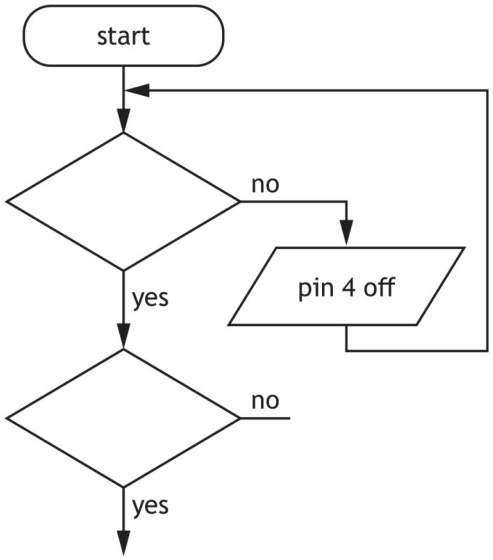
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MARKS | DO NOT WRITE IN THIS MARGIN

12. (continued)

(a) Complete the flowchart for the sequence shown opposite, with reference to the data booklet and input/output connections.

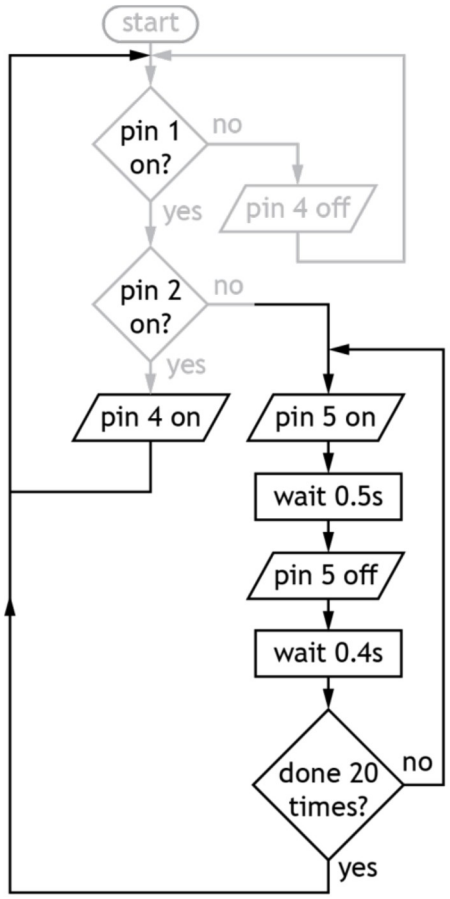
Include **all** pin numbers and the delay unit in your flowchart. **10**



[Turn over

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2026 Question 12 (a): marking instructions

Question			Expected response	Max mark	Additional guidance
12.	(a)			10	Each decision must include a question mark. If 1/0 is used rather than on/off, then an equal sign is required. If flowchart is redrawn then the yes/no routes and arrows must match that given. **1 mark** - Pin 1 on? decision in correct position. **1 mark** - Pin 2 on? decision in correct position. **1 mark** - Pin 4 on in correct position. **1 mark** - loop after pin 4 on back to start with arrow. **1 mark** - Pin 5 on **and** off in correct position. **1 mark** - 0.5s **and** 0.4s delay with Unit. **1 mark** - x 20? decision with Y/N. **1 mark** – decision no route with arrow to before pin 5 on. **1 mark** – decision yes route with arrow back to start. **1 mark** - all drawn symbols correct. Ignore any additional steps.

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2026 Question 13 (e)

N5 Engineering Science 2026 paper, Section 2, Question 13 (e). Total: 2 marks

The air source heat pump is operated by a microcontroller.

(e) Describe two advantages of **using** a microcontroller compared to a hard-wired electronic circuit in the operation of the air source heat pump. **2**

1. _____

_____

2. _____

_____

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2026 Question 13 (e): marking instructions

Question			Expected response	Max mark	Additional guidance
	(e)		Air source heat pump will be more reliable with a microcontroller (less components that could fail). It would be easier to fault find with a microcontroller (as an error code could be displayed). The air source heat pump control software can be reprogrammed/ updated (to improve efficiency). A hardwired electronic control circuit would be more difficult to update (remotely).	**2**	Descriptive response must relate to a positive advantage of microcontroller use in the operation of the air source heat pump. **1 mark** for each description. Do not accept description of initial programming of microcontroller. Marks can be awarded for maintenance related advantages but not installation/manufacture. Do not accept easier to replace parts on its own.

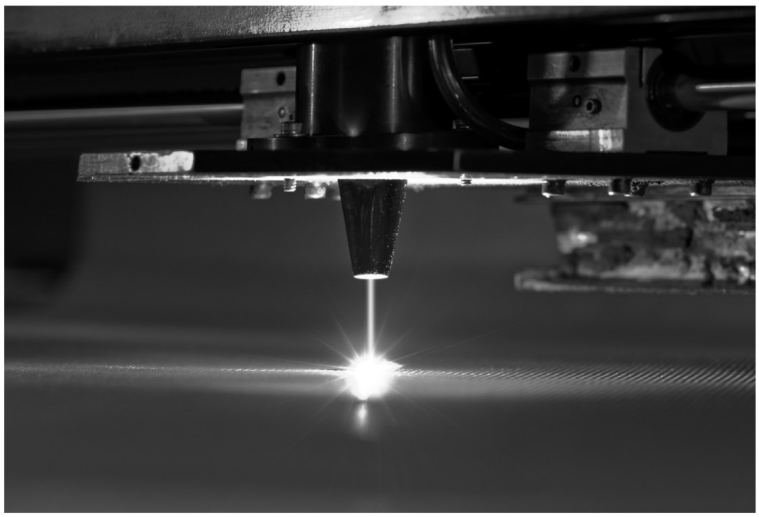
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Specimen Question 13 (a)

N5 Engineering Science Specimen paper, Section 2, Question 13 (a). Total: 7 marks

13. A laser cutting machine is used to cut sheet steel, as shown in the photograph below.

DO NOT WRITE IN THIS MARGIN



The laser is positioned by motors A and B which are operated by a microcontroller.

Input and output connections to the microcontroller are shown in the table below.

Input connection	Pin	Output connection
	7	Motor A forward (move right)
	6	Motor A reverse (move left)
	5	Motor B forward (move forward)
	4	Motor B reverse (move back)
	3	Laser (on when high, off when low)
	2	
	1	
	0	

The cutter is required to perform the following sequence of operations:

1: Switch on the laser

2: Move right for 0·5 seconds

3: Move forward for 0·5 seconds

4: Repeat steps 2 and 3 four times

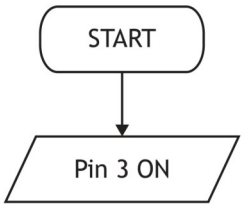
5: Switch off laser and motors

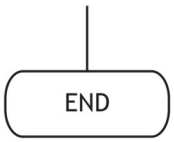
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MARKS | DO NOT WRITE IN THIS MARGIN

13. (continued)

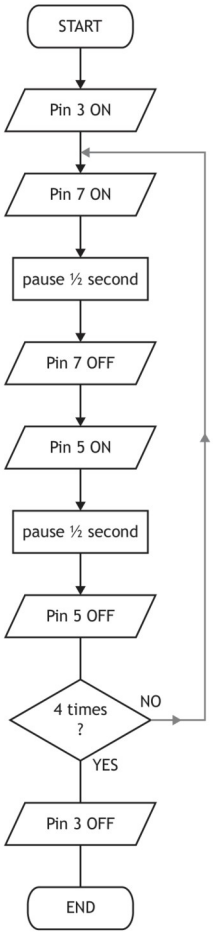
(a) Complete, with reference to the data booklet and input/output connections, the flowchart for the control sequence. Include **all** pin numbers. **7**





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Specimen Question 13 (a): marking instructions

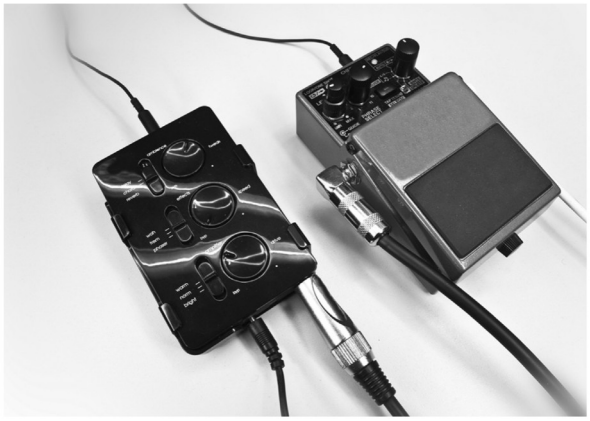
Question			Expected response	Max mark	Additional guidance
13	a			7	1 mark for all symbols used correctly 1 mark for feedback loop with arrows 1 mark for decision including yes/no 1 mark for both pause commands 1 mark for PIN 3 ON and OFF 1 mark for PIN 7 ON and OFF 1 mark for PIN 5 ON and OFF

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Specimen Question 14 (d)

N5 Engineering Science Specimen paper, Section 2, Question 14 (d). Total: 2 marks

An "effects pedal", as shown in the photograph below, can be attached to the guitar to change the sound produced. The effects pedal uses complex electronic circuitry.



(d) Explain why a microcontroller might be used in place of a hard wired circuit in the effects pedal. **2**

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Specimen Question 14 (d): marking instructions

Question			Expected response	Max mark	Additional guidance
14	d		Any valid point with explanation, such as: • reduced size • quicker assembly • reduced cost, etc … due to fewer components	2	

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